Full Stack Developer Guide

1. Who is a Full Stack Developer?

A Full Stack Developer designs, builds, tests, deploys, and maintains web applications across both frontend and backend. The role focuses on creating complete applications using modern user interfaces, APIs, databases, and cloud/deployment practices.

2. Responsibilities

• Build responsive web interfaces
• Develop backend APIs and business logic
• Design and manage databases
• Integrate frontend and backend systems
• Implement authentication and authorization
• Test, debug, and improve applications
• Deploy applications to production
• Monitor reliability, performance, and security
• Collaborate with product, design, and engineering teams

3. Required Skills

Programming

• JavaScript/TypeScript
• Python or another backend language
• HTML
• CSS

Frontend

• React
• Responsive Design
• State Management
• REST API Integration

Backend

• Node.js/Express or FastAPI/Django
• REST APIs
• Authentication
• API Validation

Databases

• PostgreSQL
• MySQL fundamentals
• SQL
• Database Design
• ORM concepts

DevOps
• Git
• Docker
• CI/CD basics
• Linux fundamentals

Cloud
• AWS, Azure, or Google Cloud

4. Preferred Skills
Next.js, TypeScript, Redis, GraphQL, WebSockets, microservices, message queues, caching, testing frameworks, system design, security, performance optimization, and monitoring.

5. Programming Languages
JavaScript/TypeScript, Python, SQL, HTML, and CSS, with optional exposure to Java, Go, C#, or PHP.

6. Frameworks
React, Next.js, Node.js/Express, FastAPI, Django, Spring Boot, and other modern frontend or backend frameworks.

7. Tools
VS Code, Git, GitHub, Docker, PostgreSQL, MySQL, Postman, npm, browser developer tools, Linux, Redis, and cloud services.

8. Learning Roadmap

Phase 1 – Web Foundations
HTML → CSS → JavaScript → Git

Phase 2 – Frontend Development
React → Components → State Management → API Integration

Phase 3 – Backend Development
Node.js/Express or FastAPI/Django → REST APIs → Authentication → Validation

Phase 4 – Databases
SQL → PostgreSQL → Database Design → ORM

Phase 5 – Full Stack Integration
Frontend → Backend APIs → Authentication → Database → Testing

Phase 6 – Deployment

Docker → CI/CD → Cloud → Monitoring → Production Optimization

9. Beginner Projects
Personal portfolio, responsive landing page, to-do application, weather application, blog application, and simple CRUD application.

10. Intermediate Projects
E-commerce application, job portal, expense tracker, real-time chat application, task management system, and full-stack authentication application.

11. Advanced Projects
Scalable e-commerce platform, real-time collaboration application, multi-tenant SaaS platform, microservices-based application, production-grade social platform, and distributed full-stack system.

12. Resume Optimization Tips
Emphasize measurable impact, full-stack projects, deployed applications, GitHub portfolio, categorized technical skills, API and database experience, cloud deployment, testing practices, concise achievements, and clean formatting.

13. Interview Topics
JavaScript, TypeScript, React, HTML/CSS, frontend architecture, REST APIs, authentication and authorization, backend architecture, SQL, database design, HTTP, Git, Docker, cloud deployment, system design, security, performance optimization, and behavioral questions.

14. Career Progression
Typical path: Student → Junior Full Stack Developer → Full Stack Developer → Senior Full Stack Developer → Lead Developer → Software Architect/Engineering Lead. Focus on progressively larger applications, stronger frontend and backend architecture, database expertise, cloud deployment, system design, security, and measurable business impact.